LIMIT TOPOLOGY EQUALS THE I-ADIC TOPOLOGY UNDER GRADED ACC

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Open mapping lemma. More on Algebra, Lemma more-algebra-lemma-open-mapping. The continuous map u is open, hence surjective.

Source reference. Original reference 0CQW: sources/stacks-chow/tags.txt, source lines 15976–15976. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let I be an ideal of a ring A. Let X be a topological space. Let*

$$\ldots \to \mathcal{F}_3 \to \mathcal{F}_2 \to \mathcal{F}_1$$

be an inverse system of sheaves of A-modules on X such that $\mathcal{F}_n = \mathcal{F}_{n+1}/I^n\mathcal{F}_{n+1}$. Let $p \geq 0$. Assume

$$\bigoplus\nolimits_{n\geq 0} H^p(X, I^n\mathcal{F}_{n+1})$$

satisfies the ascending chain condition as a graded $\bigoplus_{n\geq 0} I^n/I^{n+1}$-module. Then the limit topology on $M = \lim H^p(X, \mathcal{F}_n)$ is the I-adic topology.

Proof. Set $F^n = \mathrm{Ker}(M \to H^p(X, \mathcal{F}_n))$ for $n \geq 1$ and $F^0 = M$. Observe that $IF^n \subset F^{n+1}$. In particular $I^nM \subset F^n$. Hence the I-adic topology is finer than the limit topology. For the converse, we will show that given n there exists an $m \geq n$ such that $F^m \subset I^nM$[1]. We have injective maps

$$F^n/F^{n+1} \longrightarrow H^p(X, \mathcal{F}_{n+1})$$

whose image is contained in the image of $H^p(X, I^n\mathcal{F}_{n+1}) \to H^p(X, \mathcal{F}_{n+1})$. Denote

$$E_n \subset H^p(X, I^n\mathcal{F}_{n+1})$$

the inverse image of F^n/F^{n+1}. Then $\bigoplus E_n$ is a graded $\bigoplus I^n/I^{n+1}$-submodule of $\bigoplus H^p(X, I^n\mathcal{F}_{n+1})$ and $\bigoplus E_n \to \bigoplus F^n/F^{n+1}$ is a homomorphism of graded modules; details omitted. By assumption $\bigoplus E_n$ is generated by finitely many homogeneous elements over $\bigoplus I^n/I^{n+1}$. Since $E_n \to F^n/F^{n+1}$ is surjective, we see that the same thing is true of $\bigoplus F^n/F^{n+1}$. Hence we can find r and $c_1, \ldots, c_r \geq 0$ and $a_i \in F^{c_i}$ whose images in $\bigoplus F^n/F^{n+1}$ generate. Set $c = \max(c_i)$.

For $n \geq c$ we claim that $IF^n = F^{n+1}$. The claim shows that $F^{n+c} = I^nF^c \subset I^nM$ as desired. To prove the claim suppose $a \in F^{n+1}$. The image of a in F^{n+1}/F^{n+2} is a linear combination of our a_i. Therefore $a - \sum f_ia_i \in F^{n+2}$ for some $f_i \in I^{n+1-c_i}$. Since $I^{n+1-c_i} = I \cdot I^{n-c_i}$ as $n \geq c_i$ we can write $f_i = \sum g_{i,j}h_{i,j}$ with $g_{i,j} \in I$ and

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[1] In fact, there exist a $c \geq 0$ such that $F^{n+c} \subset I^nM$ for all n.

$h_{i,j}a_i \in F^n$. Thus we see that $F^{n+1} = F^{n+2} + IF^n$. A simple induction argument gives $F^{n+1} = F^{n+e} + IF^n$ for all $e > 0$. It follows that IF^n is dense in F^{n+1}. Choose generators $k_1, \ldots, k_r$ of I and consider the continuous map

$$u : (F^n)^{\oplus r} \longrightarrow F^{n+1}, \quad (x_1, \ldots, x_r) \mapsto \sum k_i x_i$$

(in the limit topology). By the above the image of $(F^m)^{\oplus r}$ under u is dense in F^{m+1} for all $m \geq n$. By the open mapping lemma (More on Algebra, Lemma 0CQW) we find that u is open. Hence u is surjective. Hence $IF^n = F^{n+1}$ for $n \geq c$. This concludes the proof. $\square$